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AI-Powered Competency Alignment: Self-Enforcing Networks as Strategic Tools for Digital Transformation

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Abstract

This article presents a structured, competency-based methodology for aligning employees' skills with the company's strategic goals during digital transformation. The approach addresses key challenges in AI-driven Human Resource Management (HRM), such as complexity, transparency, and regulatory compliance (e.g., EU AI Act). Utilizing a self-organized learning neural network, the Self-Enforcing Network (SEN), the methodology enables systematic assessment and visualization of individual and team competencies, supporting objective, data-based HR decisions. The methodology was designed by a real-world application and demonstrates how it can support team composition, highlight skill gaps and facilitate strategic alingment. The developed model remains adaptable to differnet organizations and organizational contexts offering a robust, reproducible solution for navigating the complexities of AI adoption in HRM while meeting the highest standards of ethical and regulatory integrity.

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Keywords: Artificial Intelligence (AI), Digital Transformation, Competency Management, Human Resource Management (HRM), Self-Enforcing Networks (SEN), Strategic Alignment

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1. Introduction: Navigating Digital Transformation. A competency-based approach to HRM and AI adoption

Digital transformation compels organizations to enhance agility and adapt to rapidly changing environments [1]. It involves substantial restructuring of business processes, cultures and business models. Competitive advantage increasingly stems from data-driven strategies and targeted AI use, enabling disruptive business models and driving market value [2] [3] [4].

Success depends not only on technology, but also on leadership and employee commitment [5]. Many initiatives fail due to cultural misalignment or weak engagement. Sustainable transformation requires active participation across all levels, reducing resistance, preserving momentum and fostering shared responsibility.

To support this, leadership competency frameworks emphasize adaptability, strategic thinking and the interplay of individual, social and environmental factors [6] [7] [8] [9]. They emphasize personality traits, competence portfolios and the specific capabilities required to navigate digital change. A strong leadership and employee empowerment are equally vital for successful transformation. Addressing both increases the likelihood of lasting outcomes. While leadership buy-in is essential, the engagement and empowerment of employees are equally critical to the success of transformation efforts. By addressing these aspects holistically, organizations can better navigate the challenges of transformation and increase the likelihood of achieving lasting positive outcomes.

This necessitates a culture of continuous learning, effective knowledge management systems, and a long-term commitment to employee development [10]. In conclusion, digital transformation demands active engagement from all organizational levels [11]. Leaders must develop dynamic capabilities to identify opportunities and manage threats in this evolving landscape [12].

Simultaneously, employees are expected to build a comprehensive understanding of emerging technologies and their strategic application within the organization [13] [14]. Meeting these demands necessitates a strategic realignment of HRM, both to address the growing shortage of skilled workers and to ensure the effective integration of AI into key business processes [15] [16] [17] [18]. The use of AI in HRM, however, is subject to increasing regulatory scrutiny, particularly in Europe, where the AI Act classifies AI systems used for recruitment or performance evaluation as “high risk” (AI Act ANNEX 3 (4)), requiring transparency, reproducibility and reliability.

This article proposes a methodological approach leveraging employee´s self-assessments to enable transparent, comprehensible competence evaluation through the Self-Enforcing Network (SEN). This fosters trust, facilitates the assembly of high-performing teams and effectively supports companies on their digital transformation journey.

To concretize the method, section two highlights key aspects of digitalization, section three describes the SEN methodology, section four presents self-assessment results and the concluding section discusses practical implications.

2. Methodology

The overarching goal is to proactively develop and strengthen organizational capabilities and team structures, ensuring effective and sustainable transformation. To achieve this goal, the methodology utilizes a structured framework comprising comprehensive components and sequential phases, each designed to systematically support organizational transformation and strategic development:

- *Systematic identification and operationalization of individual evaluation criteria*: Future oriented competence profiles are defined, considering disruptive technology trends and key strategic areas.
- *Alignment with organizational objectives*: These criteria are synchronized with the organization´s vision, strategic growth and innovation priorities, and the digital transformation agenda.
- *Integration of current organizational challenges*: Existing challenges are thoroughly analyzed and embedded within the company´s contextual landscape to ensure a holistic approach.

Clear evaluation criteria align development measures with organizational needs. By comparing these criteria with corporate goals, strategic priorities, and the specific demands of digital transformation, the methodology ensures alignment with long-term objectives. This ensures a holistic transformation strategy.

2.1. Determination of skill requirements by the company

A holistic approach provides a clear overview of the individual qualifications required and the optimal team composition. This ensures that the selected competencies are both theoretically robust and practically aligned with the company´s specific needs and its strategic objectives.

The process begins by defining overarching categories that group strategically relevant skills into thematic clusters. In this case, the following three categories serve as the foundation for structuring and specifying the required competencies: Cognitive and behavioral competencies, technical and process-oriented skills and adaptive and customer-focused competencies. These categories provide the framework for identifying subordinate skills and setting clear requirements for employees. Based on these categories, the following selection details the specific competencies associated with each cluster. This breakdown allows for a clearer understanding of the skill sets required and lays the groundwork for targeted employee assessments and strategic alignment.

Cognitive & Behavioural Competencies

- *Digital Mindset*: Encompasses attitudes and behaviors that support effective use of technologies, data and AI, crucial for identifying opportunities in a data-driven environment [19] [20].
- *Personality Traits*: Includes resilience and curiosity, essential for adapting to change [21].
- *Growth Mindset*: Reflects openness to new tools, technologies, enabling navigation of transformation [19] [22].

Technical & Process-Oriented Competencies

- *Dynamic Capabilities*: Ability to reconfigure internal and external resources to remain competitive [12].
- *Technological Skills*: Cover data analysis, software development, cybersecurity, technology use [23] [24].
- *Soft Skills*: Interpersonal and communication abilities such as problem-solving, adaptability [6].
- *Agility*: Capacity to reallocate resources quickly, respond to change, enable strategic adaptability [25].

Adaptive & Customer-Oriented Competencies

- *Process Optimization*: Improves efficiency through digital methods and tools [26].
- *Innovation Management*: Dirves growth through new ideas and technologies [27].
- *Customer Orientation*: Focuses on understanding customer needs, deliverying digital solutions [16].
- *Digital Competence*: As defined by the EU, it includes confident critical and creative use of technology [28].

Fig. 1 summarizes the identified categories and their respective subcategories from the perspective of a company.

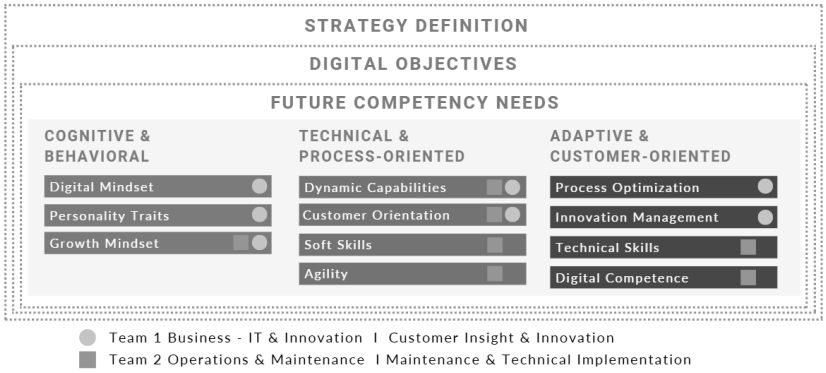

Fig. 1. Competency framework with overarching categories and assignment of main competencies per team

From a methodological standpoint, it is important that corporate goals and strategies be reformulated for employees

in the form of self-statements. This makes it easier for employees to relate to them. In consequence, a structured assessment tool was developed to capture employees´ self-evaluations. For each of the eleven defined competencies, two requirement-based statements were formulated to ensure clarity and comparability. This approach standardizes the assessment process and provides a reliable foundation for further development measures. To facilitate practical application, a 5-point Likert scale is used, allowing employees to indicate the extent to which each statement applies to them. Example statements include:

"I am open to new digital technologies in my working environment".

"I actively look for opportunities to improve my digital skills".

A maximum score of both statements indicates that the employee fully meets the requirements for the respective competencies, such as a Digital Mindset. In total, 22 statements cover the 11 competencies, enabling a comprehensive and nuanced self-assessment. This methodology not only helps identify employees who already possess the skills necessary to achieve strategic goals but also highlights areas where further development is needed. In addition, it reveals team tasks that require complementary skills, thereby supporting the optimal composition of teams for transformation initiatives. Before presenting the results, the main characteristics and advantages of the Self-Enforcing Network (SEN) will be outlined.

B. The Self-Enforcing Network

A Self-Enforcing Network (SEN) is a self-organizing method that adapts and learns from input data in an unsupervised manner [29, 30]. Like any neural network, SEN consists of input and output neurons: input neurons reflect company-defined requirements (attributes a) and output neurons represent the competencies (objects o) needed to achieve strategic goals. It includes functions, a specific learning rule and a learning rate to adjust weights. The knowledge or real data is represented in a so-called "semantic matrix" and, most importantly, the weights, which represent the connections between the neurons, are not randomly generated at the beginning.

SEN learns and clusters data according to its similarity. The according learning rule (1) "Self-Enforcing Rule" (SER) transforms the real data 1:1 into the weight matrix that contains the knowledge of a network, thus maintaining transparency and traceability. Additionally, a cue validity factor (cvf) adjusts the influence of individual attributes [29, 30].

$$\Delta w = c * w_{ao} * cvf_a \qquad (1)$$

where Δw is the change of the weight value between the attributes a and the objects o according to the value in the semantic matrix, and c is the learning rate. Several activation functions are included in SEN that return the activation value a_o of a neuron, as this function is one of the most important. In this case, the linear activation function is selected (2) which simplifies the reconstruction of the data and interpretation of the results even for non-experts.

$$a_o = \sum w_{ao} * a_a \qquad (2)$$

These characteristics of SEN thus allow the model consisting of the identified requirements and their respective competencies (see above) to be learned from the semantic matrix. Fig. 2 shows the architecture of SEN.

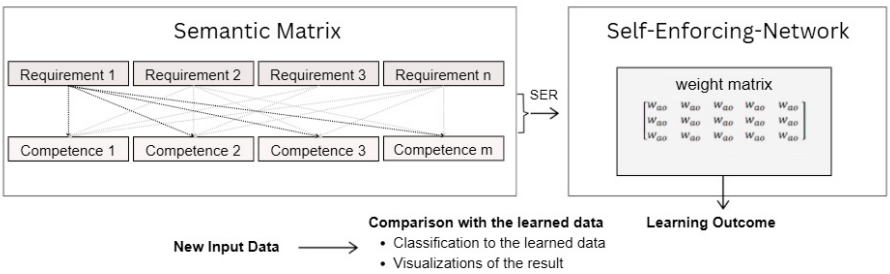

Fig. 2. Architecture of SEN

After learning, new input data is classified according to its similarity to the learned data. The methodological peculiarity, whereby employees assess the required competencies through self-assessment, is used for the input vectors. This ensures a direct mapping between the company's required competencies and the employees' self-assessments. For an input vector, the results are computed twice: as the highest activation of a neuron in the semantic matrix (displayed as "Ranking") and as the smallest distance ("Euclidean Distance"). Furthermore, the results are presented using several visualizations, as illustrated in the subsequent section.

3. Results

The comprehensive analysis reveals both frequently represented and less prevalent competencies within the organization. Identifying these patterns is crucial for refining recruitment strategies, particularly by targeting skills that are underrepresented yet essential for effective team performance. Additionally, the results provide valuable insights into individual competency profiles, enabling the strategic composition of teams in alignment with organizational objectives. This approach establishes a performance analysis framework that is dynamically tailored to the evolving strategic requirements of the organization. The following examples demonstrate the outcomes of the personal evaluation for two employees and the formation of teams.

3.1. Evaluation of competencies

Employees rate their skills and interests based on the 22 statements. These self-assessments are transferred directly to the SEN tool as input vectors. Then, these values are compared with the learned vectors from the semantic matrix to evaluate the alignment between the current self-assessments and the required skill profiles. Fig. 3 and Fig. 4 illustrate the self-assessment results for two employees, showing the computed ranking and corresponding SEN visualization.

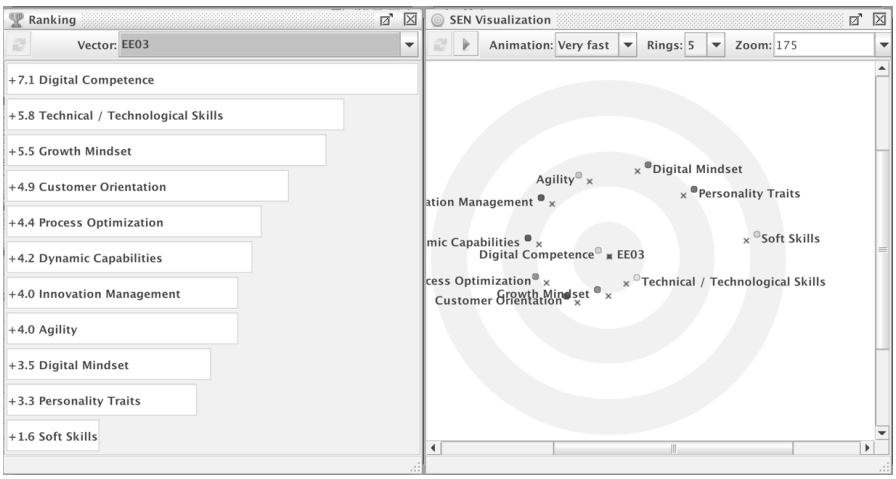

Fig. 3. SEN Visualization and the computed ranking for Employee (EE) 03

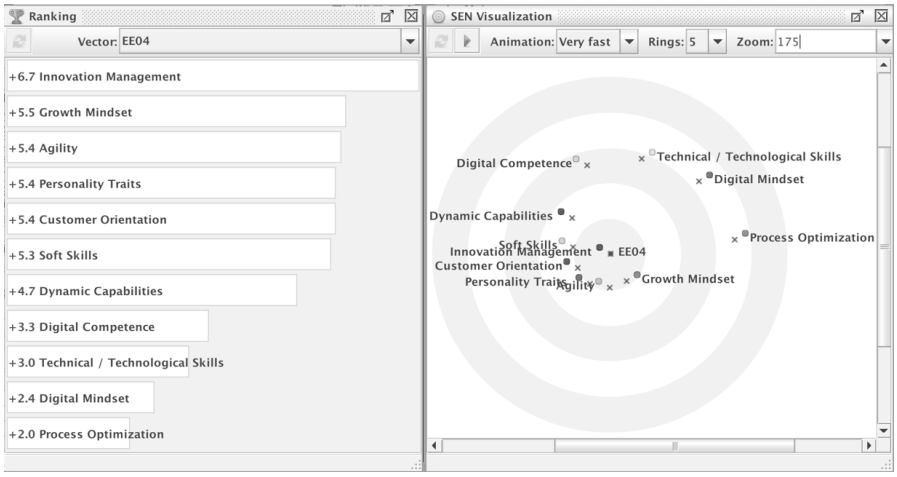

Fig. 4. SEN Visualization and the computed ranking for Employee (EE) 04

Fig. 3 illustrates the SEN visualization, placing the employee at the centre. Competency most closely aligned with the employee's self-assessment is drawn towards the centre, based on activation values calculated during the 'ranking' process. Higher activation values indicate a greater similarity between input data and the required competencies. As shown in both visualizations of Fig. 3, this employee demonstrates the highest activation for digital competences, followed by technical/technological skills and growth mindset. In contrast, competencies such as soft skills, personality traits, and digital mindset reveal areas for targeted training or development initiatives. The results shown in Fig. 4 for the Employee EE04 are to be interpreted accordingly.

3.2. Formation of teams

To support transformation, the organization defines comopetency requirements for team formation, illustrated by two teams: T1 (“Team Business – IT and Innovation”) and T2 (“Team Operations and Maintenance”).

Each team receives high baseline values for core requirements. While all employees need basic knowledge, some teams require deeper expertise in specific comptencies. For Team T1, prioritized competencies include Soft Skills, Agility, Personality Traits, Innovation Management, and Dynamic Capabilities. In contrast, Team T2 emphasizes Process Optimization, Technical/Technological Skills, Digital Competence, and Digital Mindset. Required competencies and self-assessments of six employees (EE01-EE06) are given also as input vectors. Fig. 5 displays the results by the Map Visualization according to the computed distances.

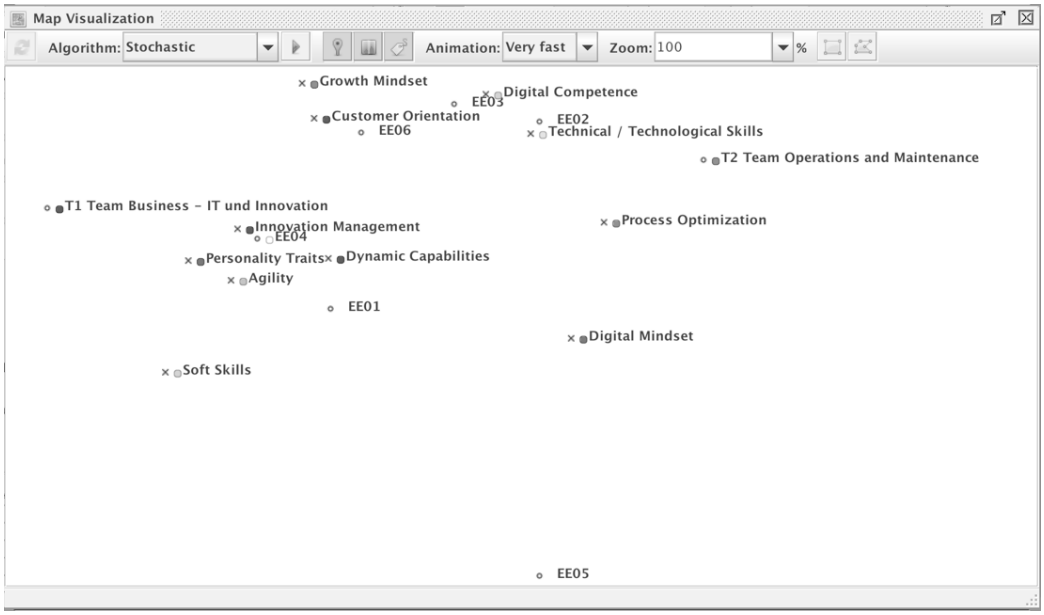

Fig. 5. Map-Visualization according to the computed distances.

In Fig. 5, Team 1 appears on the left, with competencies and employees (EE01, EE04) nearby. A gap remains as several competencies are rated at maximum, which no employee currently fulfils.

Employee coded as EE01 is located close to "Dynamic Capabilities", indicating a high proficiency in this area. EE04 shows an even stronger connection to “Innovation Management,” which is particularly relevant for digital transformation, and demonstrates strengths in “Agility” and “Personality Traits”. These findings are consistent with the self-assessment results in Fig. 3 and 4. The differing strengths between employees complement each other within the team. The visualization also highlights that soft skills are currently underrepresented, indicating a need for targeted development in this area.

Team T2 is positioned on the opposite side of the map visualization, with the relevant core competencies grouped around the team. Employees EE02 and EE03 are particularly well suited for this team, as both demonstrate pronounced “Technical/Technological Skills” and “Digital Competences”, key requirements for managing and advancing a digital business ecosystem.

Employee EE06 is located approximately midway between the two teams. Thanks to pronounced “Customer Orientation” and solid understanding of both domains, EE06 emerges as a promising candidate for the role of a mediator role, effectively bridging the gap between teams. In contrast, EE05 is positioned further from both teams and the required competencies, indicating that this employee currently lacks sufficient qualifications for assignment to either team.

These examples illustrate how self-assessments and team requirements can be integrated within a single, transparent, and reproducible model. The visualization not only supports objective team formation but also highlights areas where targeted development or training initiatives are needed to close existing competency gaps.

4. Discussion

To ensure that digital transformation efforts are strategically aligned with corporate objectives, this approach introduced a structured approach for aligning workforce competencies with organizational strategy, leveraging the Self-Enforcing Network (SEN). The approach emphasizes transparency, objectivity and contextual relevance, setting itself apart from conventional tools and aligning with the ethical guidelines of the AI Act. In terms of key findings and contributions, the approach demonstrates multiple strengths.

4.1. Key findings and contributions

The methodology enables a systematic assessment of competencies by breaking down job descriptions into specific capabilities, fostering a shared understanding of strategic direction. The evaluation process within the SEN tool ensures end-to-end transparency and data integrity, avoiding algorithmic manipulation and supporting trustworthy HR decision-making. The model maintains consistency throughout, effectively linking organizational strategy with digital transformation, recruitment and employee development. SEN based visualizations offer a clear overview of existing competencies and development needs, while direct access to unaltered data ensures traceability and integrity. Overall, the approach provides a solid foundation for AI-supported HR processes by promoting fairness, transparency and accountability.

4.2. Stakeholder Benefits

HR management gains the ability to develop precise, strategy-driven job descriptions, conduct objective candidate selection and design targeted development initiatives. Executives and managers benefit from improved strategic and operational alignment, using the approach to form high performing teams and close skill gaps across the organization. The consistent integration of strategy, transformation goals, recruitment and development enable more holistic leadership. Employees experience greater clarity around corporate objectives and personal development opportunities. By linking individual competencies to strategic goals, the approach increases employee engagement, reduces resistance to change and supports both personal growth and organizational adaptability.

Using the SEN this iterative process identifies critical skill and competency gaps for successful transformation. By linking these gaps directly to strategic objectives, SEN enables targeted interventions, such as upskilling initiatives or the redesign of cross-functional teams, to systematically enhance the organization´s adaptability to technological change. The overarching goal is to proactively develop and strengthen organizational capabilities and team structures, ensuring effective and sustainable transformation.

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